

FruitGaussian: 3D Gaussian Splatting for Automated Fruit Counting in Natural Orchard

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Abstract: Automatic fruit counting in natural orchard environments remains challenging due to foliage occlusion, illumination variations, and inefficient 3D reconstruction methods. To address these limitations, we propose FruitGaussian, a novel 3D Gaussian Splatting (3DGS)-based framework for high-fidelity orchard scene understanding and fruit counting. Our approach introduces two key innovations: a hierarchical semantic embedding mechanism that fuses multimodal features (RGB, semantics, depth) for precise fruit instance encoding, and a depth-aware regularization constraint that leverages monocular depth priors to suppress boundary artifacts and floating noise. By integrating explicit 3D Gaussian representations with differentiable rendering, FruitGaussian achieves real-time high-fidelity rendering and geometric fidelity while enabling direct point cloud extraction within seconds. Evaluations on real-world persimmon orchard datasets demonstrate a 96% counting accuracy, surpassing Neural Radiance Fields (NeRF)-based methods by 16 %. The explicit Gaussian representation enables efficient point cloud extraction within seconds, significantly advancing the practicality and robustness of vision-based yield estimation in precision agriculture.

Key Words: 3DGS, Fruit Counting, Depth Regularization, Precision Agriculture

1 Introduction

Automatic acquisition of orchard phenotypic data and yield information constitutes a fundamental requirement for smart orchard construction. With increasing global food demand, agricultural resource optimization based on advanced sensing technologies has emerged as a critical direction in precision agriculture[1]. Fruit counting technology, which quantifies crop yield distribution patterns, effectively supports harvest planning and resource allocation, serving as a vital technical pathway for sustainable agricultural development.

Fruit counting in natural orchard environments faces multiple challenges: (1) Three-dimensional geometric characteristics of fruits are susceptible to interference from foliage occlusion, illumination variations, and fruit overlap; (2) Robust feature representation in complex scenes remains challenging, limiting the generalization capabilities of traditional vision methods. Existing approaches primarily follow two technical routes: 2D image processing and 3D reconstruction.

In 2D image processing methods, early studies employed color space conversion and morphological operations[2], while deep learning frameworks like YOLO and R-CNN significantly improved performance through end-to-end feature learning[3]. Video-based methods utilize Kalman filters for multi-object tracking[4], yet suffer from tracking drift under occlusion due to inherent 2D projection information loss.

3D reconstruction methods enhance counting accuracy through multi-view geometric constraints, exemplified by Structure from Motion (SfM) and multi-view stereo vision[5–7]. However, current approaches exhibit notable limitations: (1) Dependency on expensive sensors (e.g., stereo cameras, LiDAR) for depth acquisition; (2) The sparse point clouds generated by SfM cannot adequately support fine-grained phenotypic analysis or the operation of picking robots.

Recent advances in novel view synthesis using Neural Radiance Fields (NeRF) [8] have opened new possibilities for

3D orchard scene reconstruction. As a pioneering work in this domain, FruitNeRF [9] establishes an implicit neural representation framework that integrates multi-view images to construct semantic-aware NeRF models. Through uniform voxel sampling and subsequent clustering algorithms, this approach achieves fruit counting with explicit point cloud generation. Compared with traditional SfM methods [10], This framework effectively mitigates error propagation from 2D semantic segmentation. Nevertheless, two critical limitations persist: The volume rendering paradigm of NeRF requires dense ray sampling and global optimization, leading to prohibitive computational costs that impede real-time deployment. Furthermore, inherent limitations in unbounded scene modeling tend to generate background artifacts and floaters, while the current single semantic constraint framework proves inadequate for suppressing such complex noise interference.

In this work, we present FruitGaussian, a 3D Gaussian Splatting (3DGS)-based framework for high-fidelity orchard scene understanding and automated fruit counting. The main contributions are as follows: (1) We propose the first explicit orchard modeling framework using 3DGS, enhanced by hierarchical semantic embedding mechanisms that enable effective multi-modal feature fusion and encoding fruit instances. (2) We develop a depth-aware regularization constraint that introduces implicit geometric priors to resolve boundaryless reconstruction challenges, effectively suppressing floating artifacts and ghosting phenomena while maintaining reconstruction fidelity under complex illumination conditions. To validate the efficacy of the FruitGaussian framework, we conducted systematic experimental evaluations on the dataset collected from real orchard environments. The results demonstrate that our method outperforms existing approaches across core metrics including photorealistic 3D rendering quality, point cloud geometric fidelity, and fruit counting accuracy.

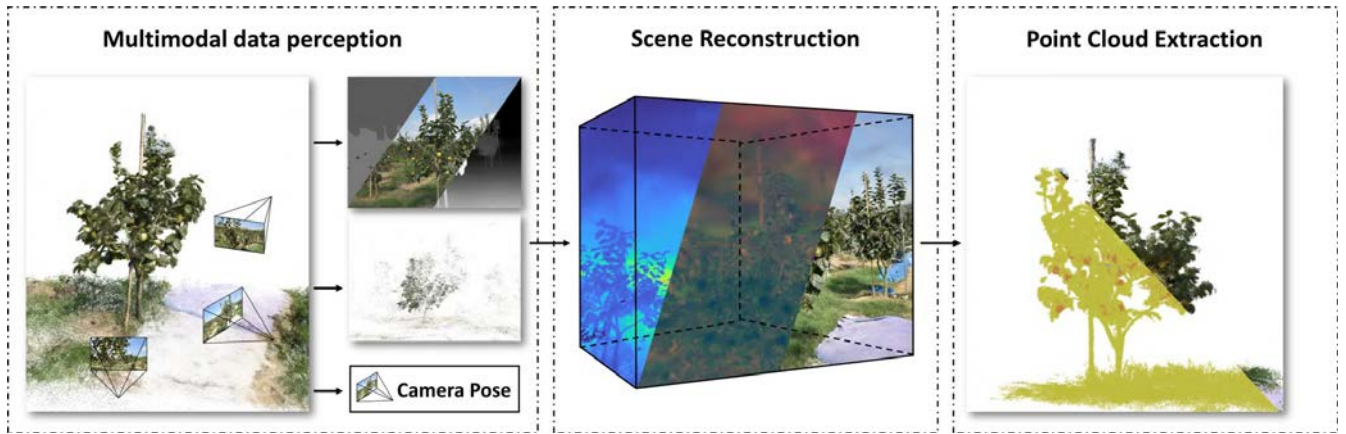


Fig. 1: **Pipeline of our proposed fruit counting framework-FruitGaussian.** **Multimodal Data Perception**(Sec. 2.1): Following the acquisition of camera poses and sparse point clouds via SfM [10], Grounded SAM2 [11][12] generates fine-grained semantic masks, which are fused with depth estimation maps from Depth Anything V2 [13] to construct multimodal scene representations. **Scene Reconstruction**(Sec. 2.2): These representations are then incorporated into a differentiable 3DGS system by injecting semantic attributes and imposing depth constraints, thereby reconstructing the 3D orchard scene. **Point Cloud Extraction**(Sec. 2.3): The resulting Gaussian point clouds undergo semantic decoupling, yielding 3D spatial fruit distributions that enable precise instance-level counting through clustering.

2 Methodology

In this section, we present the framework and pipeline of our proposed method, as illustrated in Fig. 1.

2.1 Multimodal Data Perception

The data collection was conducted in September, the optimal harvest season for persimmon maturation, at a 'Taishuu' persimmon orchard located in Feidong County, Hefei City, Anhui Province, China (geographic coordinates: 31°59' N, 117°51' E). The target cultivar exhibits phenotypic complexity with yellow-green pericarp spectrally overlapping foliage (Fig. 2), presenting significant detection challenges under dense fruit distribution and severe occlusion.



Fig. 2: **Orchard environment.** Yellow-green fruits exhibit overlap with foliage under conditions of dense occlusion and variable ambient lighting.

We used a Nikon D750 camera (equipped with a 24mm lens, set at f/8 aperture) to capture 400 RGB images (with a resolution of 4000×6000 pixels) through multi-view acquisition: 12 azimuthal positions within a 1-meter radius, each with 3 vertical tiers (spaced 0.5 meters apart). COLMAP[10] recovered camera poses ($\mathbf{K} \in \mathbb{R}^{3 \times 3}$, $[\mathbf{R}|\mathbf{t}] \in SE(3)$) and

sparse point clouds for 3DGS initialization.

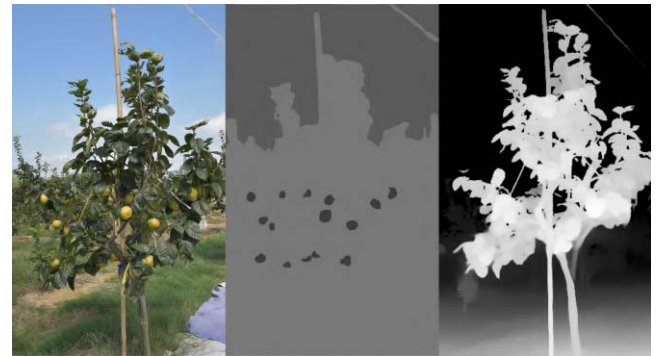


Fig. 3: **Multimodal data processing results.** Visualization of input data modalities: RGB images (left), fine-grained semantic masks (middle) from Grounded SAM2[11], and scale-aligned depth estimation maps (right) derived from Depth Anything V2[13].

Subsequently, we integrated the open-domain detection capabilities of Grounding DINO[11] with the zero-shot segmentation advantages of Segment Anything V2[12] to generate fine-grained semantic masks. Concurrently, we utilized Depth Anything V2[13] to extract monocular depth priors and employed SfM point clouds for global depth alignment. The resulting RGB images, semantic masks, and depth estimation maps are illustrated in Fig. 3. We determined the actual total number of fruits by manually counting them.

2.2 Scene Reconstruction

Traditional 3D scene representations such as point clouds, voxels, and shape descriptors (Signed Distance Function, SDF) exhibit limited capabilities in capturing and expressing details of complex objects or scenes. In recent years, implicit neural representations [8] have demonstrated remarkable performance in novel view synthesis by representing 3D scenes with coordinate-based neural networks. For instance, FruitNeRF [9] employs two distinct multilayer perceptron

(MLP) branches to output color and fruit feature based on NeRF's feature field refinement, followed by volumetric rendering to generate RGB images and high-dimensional feature maps. However, due to NeRF's implicit radiance field representation, it requires computationally intensive ray sampling during training and rendering, leading to substantial optimization costs. Moreover, implicit representations cannot directly generate explicit point clouds, significantly limiting NeRF's applicability in downstream screening and editing tasks.

Our work aims to construct an expressive 3D scene representation for natural orchard environments that achieves high-fidelity modeling of scene appearance and geometry while effectively distinguishing fruit features to facilitate downstream screening and counting tasks. We design our approach based on the recent 3DGS technique [14], extending it from pure 3D reconstruction to fine-grained scene understanding and fruit counting. The network architecture of FruitGaussian is illustrated in Fig. 4.

2.2.1 Preliminaries: 3D Gaussian Splatting

In 3DGS, a scene is represented by a set of differentiable 3D Gaussians. Each Gaussian is characterized by its **centroid** $p = \{x, y, z\} \in \mathbb{R}^3$, **covariance matrix** $\Sigma \in \mathbb{R}^{3 \times 3}$, **opacity** $o \in \mathbb{R}$ and **color** $c \in \mathbb{R}^3$ represented by third-order spherical harmonics coefficients. To ensure the positive semi-definiteness of the covariance matrix Σ during parameter optimization, Σ is decomposed into a rotation matrix $R \in \mathbb{R}^{3 \times 3}$ and a scaling matrix $S \in \mathbb{R}^3$ following Eq. 1. These matrices are further parameterized as a **scaling vector** $s \in \mathbb{R}^3$ and a **quaternion** $q \in \mathbb{R}^4$ for independent optimization.

$$\Sigma = RSS^T R^T \quad (1)$$

The tunable parameters of each Gaussian are collectively denoted as S_{Θ_i} , where $S_{\Theta_i} = \{p_i, s_i, q_i, o_i, c_i\}$ represents the set of learnable parameters for the i -th Gaussian. In Gaussian Splatting, these 3D Gaussian functions are projected onto the 2D image plane and rendered differentially per pixel through a rasterizer. The projected 2D Gaussians are depth-sorted based on their z -coordinates, with transparency values composited using the discretized volumetric rendering equation. This process ultimately yields the pixel color $\hat{C}$ formalized in the rendering equation Eq. 2:

$$\hat{C} = \sum_{i \in N} c_i \alpha_i T_i \quad (2)$$

where T_i denotes the cumulative transmittance at pixel location p , expressed as:

$$T_i = \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (3)$$

and α_i represents the Gaussian blending coefficient defined by:

$$\alpha_i = o_i \cdot \exp \left(-\frac{1}{2} (p - \mu_i)^T \Sigma_i^{-1} (p - \mu_i) \right) \quad (4)$$

During optimization, 3DGS employs a differentiable rendering pipeline that utilizes tile-based rasterizers to project anisotropic Gaussians and perform alpha blending, guided by multi-objective supervision combining L1 loss, structural similarity (SSIM) [15], and perceptual loss learned perceptual image patch similarity (LPIPS) [16] to drive gradient updates of Gaussian parameters.

In addition, in order to enhance geometric accuracy and scene adaptability, 3DGS adopts an adaptive density control strategy to dynamically adjust the generation and deletion of Gaussians.

2.2.2 FruitGaussian

This section delineates the core algorithmic design for orchard scene understanding and semantically-aware fruit extraction. Building upon 3D Gaussian representations, we augment each Gaussian with learnable semantic attributes to enable viewpoint-invariant 3D classification, inspired by [17]. Specifically, we append a 16-dimensional semantic embedding $f \in \mathbb{R}^{16}$, as intrinsic properties to each primitive, paralleling the treatment of spherical harmonics coefficients. To counteract geometric ambiguity in unbounded regions (e.g., sky), we further implement depth-aware regularization that constrains Gaussian distributions during optimization. The complete pipeline is illustrated in Fig. 4.

Semantic Scene Representation: Our framework decomposes natural scenes into three semantically disjoint categories: sky, fruit, and others. Each Gaussian learns probabilistic class affiliations through gradient-based optimization of its semantic embedding f . This design capitalizes on the intrinsic viewpoint invariance of 3D semantic attributes, which we formalize as DC components in spherical harmonics space.

Differentiable Semantic Rendering: Following the rasterization pipeline of [14], we implement depth-ordered composition with `top-N` overlapping primitives per pixel. The semantic vector f is then projected to 2D through Eq. 5:

$$\hat{F}_{\text{semantic}} = \sum_{i \in N} f_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (5)$$

where α_i is defined in Eq. 2. The rendered feature map $\hat{F}_{\text{semantic}}$ is subsequently decoded by a compact linear classification head to obtain per-pixel semantic labels.

Depth-Guided Geometry Regularization: To address unbounded natural scenes and mitigate artifacts caused by mispositioned splats in erroneous geometries, we introduce a depth prior to regularize the geometric structure of reconstructed point clouds. The density of SfM points depends on image quantity, making the sparse reconstruction (e.g., SfM reconstruction from 19 images creates a sparse depth map with 0.04% valid pixels on average.[18]) insufficient for direct dense depth estimation in few-shot settings. Even state-of-the-art depth completion methods struggle to bridge this significant information gap. We therefore employ Depth Anything V2[13], a pre-trained monocular depth estimation model, to generate full-resolution dense depth predictions D_{mono} . These predictions are then scale-aligned with the global SfM point clouds through Eq. 6, yielding final per-image dense depth estimates D_{dense} :

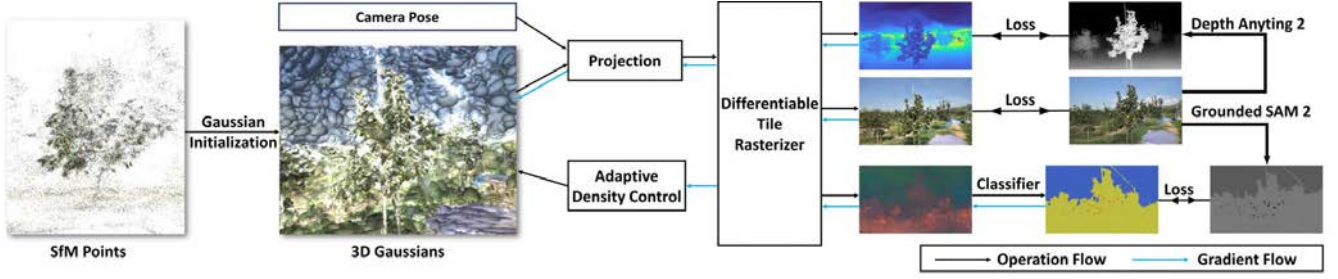


Fig. 4: **FruitGaussian network architecture.** The architecture of the proposed 3DGS-based system, illustrating the integration of semantic embeddings, depth-aware regularization, and differentiable rendering pipelines for joint appearance-geometry-semantics optimization. The proposed framework integrates three core loss functions: (a) color reconstruction loss ($\mathcal{L}_{\text{color}}$) combining structural similarity (Difference of Structural Similarity) and L1 norm constraints; (b) semantic segmentation loss ($\mathcal{L}_{\text{semantic}}$) computed as cross-entropy over differentiable rendered features after classification; (c) depth regularization loss ($\mathcal{L}_{\text{depth}}$) enforcing L1 consistency between rendered depth and precomputed dense depth maps.

$$D_{\text{dense}} = s \cdot D_{\text{mono}} + t \quad (6)$$

via Eq. 7 we adjust the scale s and offset t of estimated depth to sparse SfM depth D_{sparse} :

$$s^*, t^* = \arg \min_{s, t} \sum_{p \in D_{\text{sparse}}} \Delta(p; s, t)^2 \quad (7)$$

with $\Delta(p; s, t) = w(p) \cdot D_{\text{sparse}}(p) - D_{\text{dense}}(p; s, t)$

Depth Rendering: Our depth rendering pipeline employs a discrete volumetric formulation analogous to color rendering to compute pixel depth $\hat{D}$:

$$\hat{D} = \sum_{i \in N} d_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (8)$$

where d_i denotes view-space z-depth. For computational efficiency, we employ global depth sorting rather than per-ray ordering, with transmittance normalization ensuring physical plausibility. The shared rasterization pipeline processes RGB, depth, and features through 16×16 tiling with CUDA-accelerated pixel processing, maintaining modality-specific representations while preserving memory coherence.

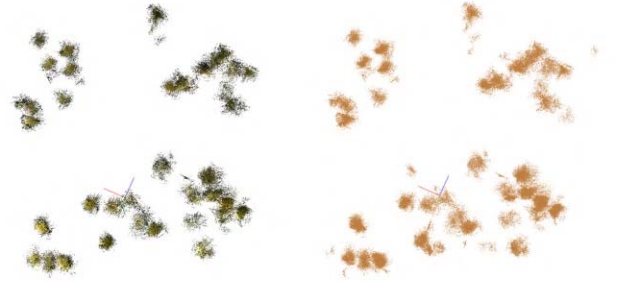
2.3 Point Cloud Extraction and Counting

Building upon the reconstructed 3D scene with semantic annotations, we develop a semantic filtering pipeline for precise fruit-oriented point cloud extraction. The proposed method directly adopts the converged Gaussian centroids from FruitGaussian training as spatial coordinates, while deriving RGB color attributes by extracting the DC components of spherical harmonics coefficients. Concurrently, a trained linear classifier decodes the semantic embeddings to assign per-point category labels. As visualized in Fig. 5, our method extracts both photometrically accurate RGB point clouds and semantically annotated counterparts.

Our fruit counting pipeline first applies semantic-color coding to annotated point clouds, linking chromatic features to category labels. It then extracts fruit-instance clusters directly via semantic-guided RGB thresholding. Subsequently, we employ Density-Based Spatial Clustering of Applications with Noise (DBSCAN)[19] for unsupervised instance



(a) RGB and semantic point clouds generated from Gaussians



(b) Fruit point clouds extracted based on semantic label

Fig. 5: **Visualization of extracted point clouds.**

segmentation. This algorithmic choice is motivated by three key merits: (1) Eliminating the need for predefining cluster counts, thus accommodating dynamic fruit distribution patterns in orchard environments; (2) Establishing density-reachable criteria that effectively discriminate between tightly packed fruit clusters and outlier noise points; (3) Preserving non-spherical cluster topology, which aligns with the natural spatial arrangement of fruit specimens.

3 Experiments and Results

To evaluate the performance of different 3D reconstruction methods in natural orchard scenarios, we conduct quantitative comparisons from both rendering quality and fruit counting accuracy perspectives. Following the benchmark protocol in [9], we adopt a 9:1 train-test split ratio across all compared methods, including parameter adjustments for FruitNeRF-big[9] to ensure fairness. Colmap reconstructions and semantic annotations share identical baselines.

Rendering quality is assessed using peak signal-to-noise ratio (PSNR), SSIM [15], and LPIPS [16] metrics.

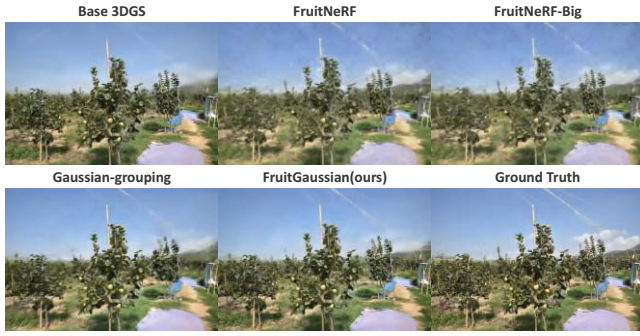


Fig. 6: **Rendering quality comparison across methods.**

All experiments were conducted on a single NVIDIA RTX 4090D with 24GB of memory. The qualitative rendering results in Fig. 7 demonstrate that our method effectively reduces artifacts and floating noise, while its detailed and accurate 3D reconstruction establishes a solid foundation for downstream point clouds editing and analysis.

This study systematically evaluates five 3D reconstruction methods on a natural orchard scene dataset. All methods are trained for 30k iterations except FruitNeRF-big, which retains its original 100k iteration setup and takes 3 hours for training.

Table 1: Quantitative Evaluation of Reconstruction in Natural Orchard Scenes

Metrics	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
FruitNeRF[9]	14.57	0.44	0.71
FruitNeRF-big[9]	14.40	0.43	0.72
Base 3DGS[14]	16.21	0.49	0.54
Gaussian Grouping[17]	16.15	0.50	0.58
Ours	16.56	0.54	0.49

As shown in Table 1, our enhanced 3DGS method achieves excellent performance: it attains a PSNR of 16.56 dB, and significantly surpasses traditional NeRF-based approaches (FruitNeRF series). Notably, our method achieves the lowest LPIPS score (0.49), demonstrating a 30.99% relative improvement over FruitNeRF variants in perceptual similarity. While maintaining comparable structural consistency (SSIM = 0.54) with Gaussian Grouping, our approach exhibits superior geometric detail preservation (PSNR) and human visual perception alignment (LPIPS). These quantitative advancements validate the effectiveness of our proposed dynamic illumination compensation and adaptive point cloud density modulation mechanisms in handling complex natural environments, establishing a robust foundation for downstream agricultural monitoring tasks such as fruit tree counting.

Fig. 7 illustrates the geometrically complete and semantically consistent point clouds generated by our approach. Notably, the explicit 3DGS representation enables point cloud exportation in approximately a few seconds, achieving two-order-of-magnitude acceleration compared to FruitNeRF's 7-minute duration.



(a) Intact fruit tree



(b) Segmented fruit

Fig. 7: **Visualization of extracted point clouds.** Qualitative comparison of reconstructed point clouds derived from our method's explicit Gaussian representation, the implicit neural fields in FruitNeRF, and the enhanced FruitNeRF-big baseline. Visual results demonstrate the geometric fidelity improvements achieved by our explicit modeling approach.

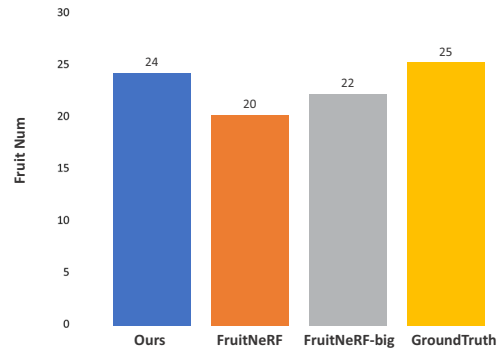


Fig. 8: **Fruit Counting Performance**

As shown in Fig. 8, the highly cohesive fruit point clouds facilitate DBSCAN clustering, yielding 96% counting accuracy that surpasses FruitNeRF baseline by 16 percentage points.

4 Limitations and Future Directions

While FruitGaussian demonstrates promising performance in orchard scene reconstruction and fruit counting, several challenges persist alongside opportunities for future exploration:

Dynamic Scene Modeling: The static scene assumption struggles to handle wind-induced vegetation movement, causing motion blur artifacts in foliage-dense regions. Incorporating temporally-aware Gaussian representations[20] could mitigate this through explicit motion modeling.

Feature-Deficient Reconstruction: Structural homogeneity and texture repetition in large-scale orchards challenge SfM with unordered images (only 30% success rate in our field tests). A hybrid approach combining 3DGS

with SLAM systems[21, 22], enhanced by multimodal sensor fusion, presents a viable solution. **Agricultural Application Directions:** Precision agriculture still requires advancements in key areas. For instance, inverse rendering techniques could enable dense geometry recovery [23] to facilitate the detection of peduncle junctions critical for robotic fruit picking.

5 Conclusion

This paper presents FruitGaussian, the first explicit 3DGS-based framework tailored for orchard scene reconstruction and automated fruit counting. By integrating depth-guided geometric regularization and hierarchical semantic embeddings, our method effectively addresses key challenges such as foliage occlusion, illumination variability, and computational inefficiency. Experimental results validate its superiority over existing approaches: FruitGaussian achieves a 96% counting accuracy (16% higher than NeRF baselines) and real-time point cloud generation (within seconds), significantly advancing the practicality of agricultural yield estimation. The explicit 3DGS representation further enables seamless downstream tasks, including semantic filtering and instance clustering via DBSCAN.

Current limitations include handling dynamic vegetation motion and texture repetition in large-scale orchards. Future work will explore spatiotemporal Gaussian representations and SLAM-3DGS fusion to enhance robustness. Our framework highlights the potential of geometry-semantics fusion for scalable precision agriculture.

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